Artificial Intelligence & Machine Learning
Task 1 Report

Project: House Price Prediction Using Linear Regression

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Dataset: California Housing Dataset

1. Introduction

This project implements a supervised machine learning solution to predict median house prices using the California Housing dataset. The project demonstrates the complete machine learning workflow including data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and model persistence.

2. Objectives

- Load the California Housing dataset using scikit-learn.
- Explore the data using summary statistics and visualizations.
- Train a Linear Regression model.
- Evaluate performance using MAE, RMSE and R^2 Score.
- Save the trained model for future predictions.

3. Dataset

The California Housing dataset contains 20,640 records with eight numerical input features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude and Longitude. The target variable is MedHouseVal (median house value). No missing values were found during preprocessing.

4. Exploratory Data Analysis

EDA included dataset inspection using head(), info() and describe(), checking for missing values and duplicates, plotting a correlation heatmap, histograms, boxplots and pair plots. These visualizations helped understand feature distributions and relationships with the target variable.

5. Model Development

The dataset was split into 80% training data and 20% testing data. A LinearRegression model from scikit-learn was trained using the training set. Predictions were generated for the testing set and compared with actual values.

6. Results

Model Evaluation Results:

Mean Absolute Error (MAE): 0.5332
Root Mean Squared Error (RMSE): 0.7456
R^2 Score: 0.5758

The model provides a reasonable baseline for predicting house prices and explains approximately 57.58% of the variance in the target variable.

7. Conclusion

The project successfully demonstrated the complete machine learning lifecycle using Linear Regression. The trained model was saved as linear_regression_model.pkl for reuse. Future work may include feature engineering and more advanced algorithms such as Random Forest, Gradient Boosting or XGBoost to improve predictive performance.

8. Technologies Used

Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn and Pickle.